



## Term Evaluation Theory (Even) Semester Regular Examination May 2026

Roll no. 2294038.....

Name of the Course: B. Tech.

Semester: VIII

Name of the Paper: Soft Computing

Paper Code: TCS-821

Time: 1.5 hour

Maximum Marks: 50

**Note:**

- (i) Answer all the questions by choosing any *one* of the sub-questions
- (ii) Each question carries 10 marks.

1 (a) Define ANN architecture and explain different types of network architectures. (CO1)  
OR

(b) Describe the building blocks of ANN and explain different activation functions with suitable examples. (CO1)

Q2. (a) Describe the components and working of human nervous system related to learning and memory. (CO1)  
OR

(b) Explain the process of setting weights in ANN and its importance. (CO1)

Q3. (a) Explain the feedback networks and describe the working of Bidirectional Associative Memory (BAM). (CO2)  
OR

(b) Describe the application procedure of perceptron with a suitable example. (CO2)

Q4. (a) Explain the Backpropagation learning algorithm step by step. (CO2)  
OR

(b) Explain the advantages and limitations of Backpropagation Network. (CO2)

Q5. (a) Explain Self-Organizing Feature Maps (SOM) with algorithm. (CO2)  
OR

(b) Describe Learning Vector Quantization (LVQ) and compare it with Self-Organizing Map (SOM). (CO2)